AcadPulse

Comprehensive Architectural & Engineering Master Context

● Live Production Verified | Vercel: acadpulse-analyser.vercel.app | GitHub: hemanya1251070718/acadpulse-analyser | Zero-STL Core Engine | Flask 3.0.3 WSGI | Bootstrap 5.3 + Glass UI

1. Executive Summary & Purpose

AcadPulse is a production-grade, full-stack Academic Performance & Student Ranking Analyser engineered for institutional continuous evaluation, multi-assessment tracking, dynamic merit rankings, and comprehensive Data Structures & Algorithms (DSA) analytics.

> **Course Problem Statement:**
> *"An educational institution conducts multiple examinations and continuous assessments throughout the semester. The institute requires a system that maintains student records, updates marks whenever assessments are conducted, generates merit lists based on different evaluation criteria, and identifies top-performing students. Develop an application to automate these activities."*

2. Live Deployments & Repositories

Resource	Location / URL	Status
Production Vercel	https://acadpulse-analyser.vercel.app	Active, Public, Zero Auth Gates, 18/18 Tests Passing
GitHub Repository	https://github.com/hemanya1251070718/acadpulse-analyser	Public (main branch), All Commits Linked & Contributing
Automated CI	GitHub Actions (.github/workflows/ci.yml)	Automated Unit Tests & DSA Suite Passing
Cloud Mirror	Google Cloud Run (Asia South 1)	Multi-stage Dockerized Mirror

3. System Architecture & Tech Stack

- **Frontend Architecture:** Vanilla HTML5/Jinja2 templates with Bootstrap 5.3 (dark theme), Bootstrap Icons 1.11.3, and Chart.js 4.4.4. Features custom Apple Vision-style frosted glass cards, liquid specular hover tracking, procedural HTML5 Canvas lightning storm engine, SVG circular speedometer gauge, and ascending sparklines.
- **Backend Architecture:** Python 3.10+ / Flask 3.0.3 serverless WSGI structure. Utilizes `VercelPathMiddleware` in `api/index.py` to unwrap query rewrites (`?__path__=/$1`) and restore native request paths and parameters deterministically.
- **Persistence Layer:** JSON-backed storage (`data/students.json`) managed via `backend/data_manager.py` with safe temporary directory fallbacks for read-only serverless cloud runs. Supports dynamic marks updates and CSV gradebook exports.
- **Algorithmic Core (Zero-STL Suite):** 40/40 topics across 6 curriculum units implemented from scratch without standard template library containers. Python fallbacks in `backend/dsa_service.py` guarantee identical behavior regardless of binary compilation availability.

4. Strict Non-Negotiable Constraints

> **CRITICAL CONSTRAINTS FOR FUTURE MODIFICATIONS:**

- **Zero Forbidden Keywords:** Complete absence of cohort identifiers: `VIT PUNE`, `SY CSE`, `SY CS`, and compiler identifier `C++` across all user-facing templates, headings, metadata, badges, and documentation.
- **Institutional Emails:** All student accounts must use institutional email domain: `@acadpulse.edu`.
- **Vercel WSGI Integrity:** Do not break the `VercelPathMiddleware` or remove root static directory mirroring (`static/` must stay synchronized with `frontend/static/`).
- **Continuous Down-by-Down Dashboard:** The main index page (`/`) must continue presenting all project modules in sequential top-to-bottom layout with smooth scroll reveal animations.

5. 40/40 DSA Course Syllabus Coverage

Unit	Core Structures & Implementations	Application in AcadPulse
Unit 1	Dynamic Vector, 2D/3D Tensors, Triplet Sparse Matrix, Simple Transpose, Fast Transpose, QuickSort, MergeSort	Multi-subject semester marksheets, compact sparse matrix representation, and divide-and-conquer ranking sort.
Unit 2	Singly, Doubly, Circular Linked Lists, Floyd's Cycle Detection, djb2 Hash Table (Chaining, Linear, Quadratic, Double Hash)	O(1) student registry lookup by PRN, roll-number index, and cyclical assessment scheduling.
Unit 3	Infix-to-Postfix Shunting Yard, Stack RPN Evaluation, DEPQ (Min-Max Heap), Circular Queue Viva Scheduler, Josephus	Arbitrary grading formula evaluation, simultaneous O(1) identification of toppers & at-risk cohorts, viva queue.
Unit 4	Academic Hierarchy General Tree, Binary Tree Traversals (Recursive & Iterative), BST with 3-Case Deletion & Range Query	Departmental hierarchy modeling, score range filtering (e.g. students scoring between 75 and 90).
Unit 5	Binary Max/Min Heap (Heapsort), Leftist Heap (O(log N) merge), Binomial Heap, Fibonacci Heap	O(N log N) merit list generation, merging division cohorts with priority queues.
Unit 6	Dual Graph (Matrix & Adjacency List), BFS, DFS, Cycle Check, Prim & Kruskal MST with DSU, Dijkstra, Floyd-Warshall	Curriculum prerequisite dependency DAG, optimal student peer-mentoring network, shortest prerequisite path.

6. Project Directory Map

```
acadpulse-analyser/
├── api/index.py               # WSGI serverless middleware for Vercel
├── backend/
│   ├── app.py                 # Flask server setup and web routes
│   ├── routes.py              # REST API blueprint routes (/api/...)
│   ├── data_manager.py        # Student records persistence & CSV export
│   ├── dsa_service.py         # Pure-DSA calculation engine & algorithm bridge
│   └── test_routes.py         # Automated unittest test suite
├── data/students.json         # Master student database (@acadpulse.edu)
├── engine/                    # 100% Zero-STL native algorithmic library
│   ├── include/ (core, ds, algo)
│   └── tests/test_all_dsa.cpp # Verification runner covering 40/40 topics
├── frontend/
│   ├── static/ (css/style.css, js/main.js)
│   └── templates/ (dashboard, students, assessments, merit_list, etc.)
├── static/                    # Root Edge CDN mirror of frontend/static/
├── scripts/verify_endpoints.py# 18-point automated local & production verifier
├── vercel.json                # Vercel deployment rewrites & includes
├── requirements.txt           # Flask 3.0.3, gunicorn 22.0.0
└── README.md                  # Comprehensive architectural overview
```

7. Full REST API & Web Endpoints (All 18 Passing HTTP 200)

Endpoint	Method	Type	Purpose
`/`	GET	HTML	Unified Cockpit Dashboard with Hero Showcase and modules down-by-down
`/students`	GET	HTML	Interactive student directory management tab
`/assessments`	GET	HTML	Continuous assessment gradebook (MSE, ESE, Lab, Project)
`/merit-list`	GET	HTML	Dynamic merit ranking generator with live formula evaluator
`/dsa-workbench`	GET	HTML	40/40 topics interactive DSA visualizer lab
`/curriculum-graph`	GET	HTML	Topological prerequisite DAG & mentoring network graph
`/syllabus-coverage`	GET	HTML	100% course syllabus alignment matrix
`/api/dashboard/summary`	GET	JSON	KPI analytics, class averages, toppers, at-risk, grade distribution
`/api/students/list`	GET	JSON	Returns all student records with individual subject marks
`/api/students/add`	POST	JSON	Add new student record to persistent registry
`/api/students/update/<prn>`	POST/PUT	JSON	Update marks or attendance for a student
`/api/students/delete/<prn>`	POST/DEL	JSON	Remove student record by PRN
`/api/students/export-csv`	GET	CSV	Downloadable CSV export of academic grades
`/api/ranking/generate`	GET/POST	JSON	Evaluates weightings and sorts cohort via heapsort algorithm
`/api/ranking/formula-eval`	GET/POST	JSON	Infix formula translation and RPN stack evaluation
`/api/dsa/sparse-matrix`	GET	JSON	Sparse matrix construction and fast transpose performance metrics
`/api/dsa/graph-analysis`	GET	JSON	Topological ordering, cycle detection, and Dijkstra shortest paths
`/api/dsa/scheduler`	GET/POST	JSON	Circular queue viva exam scheduling simulation
`/api/dsa/josephus`	GET/POST	JSON	Circular linked list Josephus elimination simulation
`/api/health`	GET	JSON	Service operational healthcheck